

Quick Check

Lesson 5: Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.3: Excretion in Plants | Lesson 5

Tea leaves from Kericho contain caffeine and miraa (khat) contains cathinone. Both are alkaloids. Which statement about alkaloids is correct?

- A** Some alkaloids are useful in small controlled doses but harmful when abused
- B** All alkaloids are harmful and should never be used
- C** Alkaloids are harmless and cannot cause addiction
- D** Only alkaloids from tobacco and miraa can harm people

A farmer cuts the stem of a Euphorbia hedge and thick milky latex oozes out and later hardens. Which excretory process released the latex?

A Organ fall

B Exudation

C Transpiration

D Guttation

In Nakuru, tanners soak animal hides in a solution made from the bark of Acacia or wattle trees. Which plant excretory product in the bark makes this possible, and what is it used for?

- A** Pyrethrum, used to make insecticide
- B** Quinine, used to treat malaria
- C** Tannins, used for tanning hides into leather
- D** Caffeine, used to flavour drinks

Early in the morning, droplets of liquid water are seen along the edges of arrowroot (taro) leaves. Which statement correctly contrasts guttation with transpiration?

- A** Guttation loses water vapour through stomata, while transpiration loses liquid droplets through hydathodes
- B** Guttation and transpiration are the same process and both happen only at night
- C** Guttation removes sticky resin from cut stems, while transpiration removes fallen leaves
- D** Guttation loses liquid water droplets through hydathodes at leaf edges, while transpiration loses water vapour mainly through stomata

A learner asks, 'Why does a plant not need a kidney?' Which answer is the best scientific reasoning?

- A** Plants produce very toxic wastes that must be kept inside the cells at all times
- B** Plants absorb all their wastes from the soil and release them through the roots
- C** Plants have low metabolic rates and less toxic wastes, recycle CO₂ and O₂ between photosynthesis and respiration, and store other wastes in dead tissues
- D** Plants do not produce any wastes

A group interviewed 12 households in their community. 7 of the 12 households said they use neem leaves to treat malaria or skin problems. What percentage of the households use neem?

A 7%

B 70%

C 41.7%

D 58.3%

In Lesson 1, a learner drew a model showing that plants have no wastes. After Lessons 1-5, which is the correct revised idea for the final model?

- A** Plants have no wastes, so nothing needs to be removed
- B** Plants produce wastes, but these have no use to people
- C** Plants produce wastes, but all of them are removed through kidneys in the roots
- D** Plants produce wastes, but they remove or store them in different ways, such as exudation, guttation, deposition and organ fall

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).